Order Perissodactyla

Horses and their relatives, tapirs, and rhinos form a group colloquially referred to as odd-toed ungulates as they have one or three functional toes on each of their hindfeet. These large herbivores have precocial, singleton offspring which suckle from inguinal mammae. Males lack a baculum. Reviews of secondary sex ratios (7407a) and reproductive cyclicity (9350) are available.

Order Perissodactyla, Family Equidae, Genus *Equus*

Horses, asses, and zebra (*Equus* 175-800 kg) are social, terrestrial grazers predominantly in Africa and Asia. The 2.5-3.5 kg, epitheliochorial, diffuse placenta, which secretes gonadotrophic hormones (78a, 198a, 198b, 751, 7059b, 9296a, 11048a), implants in a bicornuate uterus (2673, 5598, 6635, 10168, 11963), and may be eaten after birth (5868, 10168; cf 5872, 6193, 11377). Estrus usually occurs within 10 d (789, 5762, 5868, 5869, 10168, 10984, 11377; cf 11336) of the birth of a precocial, furred, and open-eyed neonate (558, 4643, 5453, 5868, 6193, 11253, 11377, 11963). Females have 2 mammae (331, 5453, 5756).

Details of anatomy (1121, 10168), endocrinology (1761, 1973b, 2201a, 5749, 5804a, 6954, 9428a), embryo transfer (10550), spermatogenesis (8356), birth (5872, 11377), and milk composition (251, 5756, 8041, 9657, 10620) are available. Supplements 23, 27, 32, and 35 of the Journal of Reproduction and Fertility, published in 1975, 1979, 1982, and 1987 respectively, contain a wealth of information on virtually all aspects of reproduction in the domestic horse, thus such is not reviewed here.

NEONATAL MASS		
Equus asinus	27-31.5 kg; n3	6193
Equus burchelli	28.4 kg; n9	5169
	30-35 kg	5756
	f: 31.5 kg; r25-35; n18	11377
	m: 33.3 kg; r28-37.5; n14	11377
	33.7 kg; se1.39; n4	10166
	34.82 kg; r31.78-39.04; n13 term emb	10166
Equus caballus	79.2 kg; sd10.0; 1 wk	2694
Equus grevyi	40 kg	5756
Equus zebra	f: 35 kg; n1	5189
WEANING MASS		
Equus burchelli	205.0 kg; se10.75; n6, 1 yr	10166
LITTER SIZE		
Equus spp	1 emb; n1 f	5052
Equus asinus	1	5365
Equus burchelli	1	11253
	1	5869
	1 emb; n1 f	6612
	1 ea; n2 lt	2109
	1 ea; n111 lt	1230
	1; n151+ births	5868
	1; 2 very rare	331
Equus caballus	1 ea; 9 lt	2109
	1.0; r1-2; 0.18% twins; n567 lt	10253
Equus grevyi	1 yg	2109
	1 ea; n13 lt	12128
Equus hemionus	1	2300
	1	4643
	1	8972
	1	558
	1 emb; n1 f	182
	1 yg ea; n24 lt	5964
Equus zebra	1	5189
	1 ea; n10 lt	1230
SEXUAL MATURITY		
Equus asinus	f: 1st preg: 1-2 yr	9367
	f: 1st birth: <2 yr	8060
	f: 1st birth: 2 yr	11908
	f/m: 2.5-3 yr	6193

Equus burchelli	f: 1st estrus: 12-15 mo	5867, 5869
	f: 1st estrus: 15.5 mo	11377
	f: 1st estrus: 18 mo; min 13	5868
	f: 1st estrus: 1.83 yr; n1	5756
	f: 1st conceive: 28 mo; r22-33	5756
	f: aborted: 2 yr 19 d	11377
	f: 1st birth: 3 yr	5867
	f: 1st birth: 3.5 yr	5868, 5869
	f: 1st birth: 3 yr 7 mo 14 d	11377
	f: 2.5 yr	7204
	m: 1st mate: 1 yr 6 mo 19 d	11377
	m: 1st mate: 5-6 yr	5868, 5869
	m: epididymal sperm: mode 3.5 yr; min 2	10169
	m: testes: large 2.5-4 yr	5756
Equus caballus	f: 1st estrus: yearling	9450
	f: 1st estrus: ~2 yr	10984
	f: 1st mate: mode 3 yr; min 2 yr	3195
	f: 1st preg: 28.75 mo; r26-31.5; n2	11286
	f: 1st birth: 2 yr	9705
	f: 1st birth: mode 3 yr; min 2	9450
	f: 2 yr	11334
	m: 1st sire: 30.5 mo; r30-31; n2	11286
	m: 31-36 mo	11334
	m/f: 2 yr; n large	3616a, 3616b
Equus grevyi	f: 3-4 yr	5756
	m: 4 yr	5756
Equus hemionus	f: 1st birth: 2-3 yr	559
	f: 1st birth: 4 yr	10896
	m: 3-4 yr	559
	f/m: 3 yr	8322
Equus zebra	f: 1st estrus: 12-13 mo	5454
	f: 1st birth: 3 yr	5454
	f: 1st birth: 4-5 yr	10158
	f: 1st birth: 53.4 ml; r46-58; n5	8355
	f: 1st birth: 66.5 mo; r38-105; n29	8355
	f: 1st birth: median 67 mo	8354
	m: 14-16 mo expelled from group, but was sexually immature	5453
	m: testes large: 3.5 yr	5454
CYCLE LENGTH		
Equus burchelli	estrus: 6 d	10168
	17-24 d	11377
Equus caballus	estrus: ~1 wk	10984
	19.7 d; n15 cycles, 6 f	394
	21 d; se0.6; n5 f	8608
Equus hemionus	estrus: 3-7 d	8933
GESTATION		
Equus asinus	11 mo	152
	r377-389 d; n15	6193
Equus burchelli	11 mo 6 d; 11 mo 20 d; n2	1355
	359-379 d	11963
	360-390 d	10158
	~1 yr	11697
	371.2 d; r361-390; n28	11377
	372 d; n18	5169
	396 d; n1; last mating to birth	10168
	11 mo 20 d-11 mo 25 d	2109
Equus caballus	320 d; (11+ mo)	11286
	337 d; se8.3; n5 before Apr 1	789
	337.2-343.3 d; across breeds	1125
	min 339 d	10984
	340 d	9705
	346 d; se8.1; n33 after Apr 1	789
Equus grevyi	406 d	345

Equus hemionus	334-358 d	8933
	~11 mo	558, 559
	~11 mo	8322
	340 d; se5	10896
	~1 yr	10207
Equus kiang	7 mo; field data	8740
	10.5 mo; field data	3034
Equus zebra	362 d; n1	5453, 5454
	12 mo	4582
	~1 yr	10158
LACTATION		
Equus burchelli	solid food: 1 wk	5762
	wean: 11 mo; r9-16	10168
	wean: 15 mo; n1	11377
Equus caballus	wean: 6 mo	10253
	wean: ~1 yr	10984
Equus grevyi	wean: 32-42 wk	5756
Equus hemionus	solid food: 1 mo	558
	nurse: ~9-11 mo	558
	wean: 1-1.5 yr	10207
Equus zebra	solid food: 3 d	5452, 5453
	reduced suckling: 2 mo	5453
	reduced suckling: 6 mo	5452
	wean: 10 mo	5452, 5453
INTERLITTER INTERVAL		
Equus burchelli	381.5 d; r378-385; n2	5867, 5868, 5869
	2 yr	11253
Equus caballus	~1 yr	789
	mode 2 yr	11336
Equus hemionus	1-2 yr	8322
Equus zebra	median 25 mo; r1-6.5 yr; n100 birth, 49 f	8354
	18 mo	10158
SEASONALITY		
Equus asinus	mate: Aug-Oct; Kutch, India	152
	mate: yr rd; CA	11908
	birth: est Mar-July; AZ	9367
	birth: July-Sept; Kutch, India	152
	birth: yr rd; n15; zoo	6193
Equus burchelli	testes large: Nov-Mar, peak Jan; S Africa	10169
	birth: 61% Jan-Mar, 85.5% Oct-Mar; Tanzania	5868
	birth: Apr-June 96%; n294; Britain	10984
	birth: May, Aug; n2; captive	2109
	birth: peak July-Sept, some Feb-Apr; Zaire	11253
	birth: July-Sept, peak Aug-Sept; n Zimbabwe	331
	birth: peak Aug-Sept; e Africa	7854
	birth: Aug-Dec; Angola	11697
	birth: Sept-Feb, peak Nov-Feb; S Africa	8514
	birth: Sept-Mar, peak Nov-Feb; S Africa	3145
	birth: Oct-Apr, peak Dec-Jan; S Africa	10167, 10168
	birth: none Aug-Sept; s Zimbabwe	2275
	birth: yr rd, 85.5% Jan-Mar; n151; e Africa	5868
	birth: yr rd, peak Oct-Mar; S Africa	10158
	birth: yr rd, many Oct-Mar, 61% Jan; Tanzania	5867, 5869
	birth: yr rd; zoo	1230
	birth: yr rd; n25; zoo	5818
	birth: yr rd; Zambia	331
	repro: yr rd, peak Dec-Apr; Tsavo East, Kenya	6413
Equus caballus	birth: mid Apr-June, peak May-mid June; WY, MT	3194, 3195
	birth: yr rd, most Apr-June, peak May; captive	11334, 11335
Equus ferus?	birth: Feb-Oct; n28; zoo	5818
Equus grevyi	mate: July-Aug, Oct-Nov; n Kenya	5762
	birth: peak Aug-Sept; e Africa	7854
	birth: yr rd?; n Kenya	5762
	birth: yr rd; n13; zoo	12128

Equus hemionus	estrus: end May-mid June; Kazakhstan	8933
	estrus: June-Sept; Kazakhstan, Mongolia	558, 559
	mate: Feb-Mar; Badchys, Turkmenia	559
	mate: mid Aug; Tibet	8972
	mate: Sept; Koko Nor, Mongolia	182
	mate/birth: mid Apr-mid May; Turkmenia	10207
	preg f: June; n1; Mongolia	182
	birth: Apr-June; Kazakhstan, Mongolia	558, 559
	birth: Apr-June; USSR	4643
	birth: May-July; n229; zoo	8322
	birth: end May-early June; n25; Ukraine, captive	10896
	birth: June-Aug, n56; zoo	5818
Equus kiang	estrus: mid Aug-mid Sept; Asia	3034
	mate: Sept; according to Mongols; Mongolia	8740
	birth: May; according to Mongols; Mongolia	8740
	birth: mid June-July; n14; zoo	12128
	birth: mid July; Asia	3034
Equus zebra	testes: large Jan, small Sept; Namibia	5454
	mate: peak Feb; Namibia	5454
	birth: June-Aug; n11; zoo	5818
	birth: yr rd, peak Nov-Apr; Namibia	5454
	birth: yr rd, peak Dec; S Africa	8354
	birth: yr rd; zoo	1230
	repro: yr rd; S Africa	10158

Order Perissodactyla, Family Tapiridae, Genus *Tapirus*

Tapirs (*Tapirus bairdii, T. indicus, T. pinchaque, T. terrestris*) are large (100-300 kg) herbivores from tropical Central and South America, and southeast Asia. Tapirs have an epitheliochorial, villous, diffuse placenta, (7563) which is eaten after birth (5789, 8990, 9673, 11352). Estrus occurs 22-35 d (11352) after the birth of a furred (3237) neonate who can walk within hours (8990). Males are ascrotal (281a). Details of endocrinology (5550) and milk composition (8124) are available.

NEONATAL MASS		
Tapirus spp	m: 6 kg; n1	12032
	8.45 kg; n1	6098
Tapirus indicus	~3 kg	3237
	m: 3.1 kg; n1	8990
	6-7 kg	6369
	6.7 kg; n1	9625
	f: 7.3 kg; n1	8990
	f: 9.40 kg; n1	9738
	m: 10.20 kg; n1	9738
Tapirus pinchaque	4.72 kg; r4.05-5.4; n2	1099
Tapirus terrestris	4.3 kg; n1	12046
	f: 4.54 kg; n1	6748
	10 kg	281a
NEONATAL SIZE		
Tapirus indicus	72 cm; n1	9625
LITTER SIZE		
Tapirus spp	1; n1	12032
Tapirus pinchaque	1 yg	1099
Tapirus terrestris	1	12128
	1 yg ea; n1 lt	6748
	1 yg ea; n9 lt	509
SEXUAL MATURITY		
Tapirus indicus	f: 1st conceive: 36 mo	8990
	f: 1st conceive: 4-4.5 yr	9738
CYCLE LENGTH		
Tapirus indicus	29.4 d; r29-31; n1 f	8990
	30 d; n1 cycle, 1 f	11352
Tapirus terrestris	64 d; r28-101; n5 cycles, 3 f	11352

GESTATION		
Tapirus indicus	9 mo; n1	3237
	13 mo	1355
	~13 mo; n1	9625
	390-395 d	6369
	401 d; r396-407; n4	8990
	403 d	9738
	405 d; n1	11352
Tapirus pinchaque	392-393 d; n1	1099
Tapirus terrestris	385-402 d	11352
	r392-405 d; n8 lt, 2 f	509
	~13 mo	1355
LACTATION		
Tapirus spp	wean: 3 mo	12032
Tapirus indicus	wean: 153 d	8990
	wean: 6-8 mo	6369
Tapirus terrestris	milk present: 9 mo	6748
INTERLITTER INTERVAL		
Tapirus indicus	2 yr	6369
SEASONALITY		
Tapirus indicus	mate: Apr-May; Thailand	6369
Tapirus terrestris	birth: Feb, May-June, Aug-Oct; n7; zoo	12128

Order Perissodactyla, Family Rhinocerotidae

Rhinoceroses (*Ceratotherium* 1400-3600 kg; *Dicerorhinus* 900-2000 kg; *Diceros* 720-1800 kg; *Rhinoceros* 1600-4000 kg) are large, terrestrial herbivores in Africa, India and southeast Asia. The 4960 g (n1), epitheliochorial, diffuse placenta (232, 2674, 6635) implants into a bicornuate uterus (3844a, 4108) and may be eaten after birth (4024, 6185, 6255, 9571). Estrus occurs from 2 wk to 8 mo (1433, 2599, 5762, 6185, 6251, 8167, 8973) after the birth of an open-eyed, mobile neonate (3983, 4024, 5504, 5762, 8167, 8798, 10118, 10158, 10867). Females have 2 teats (9155, 10158). Males have seminal vesicles, but no scrotum (3395).

Details of anatomy (3844a), endocrinology (5549, 5552, 8893), copulation (3827), birth (6255, 6943), suckling (6703), and milk composition (407, 4025, 4057, 4117, 5551, 6185) are available.

NEONATAL MASS		
Ceratotherium simum	m: est 32-36 kg; n1	6943
	~40 kg	10158
	~47.7 kg; n1, 10 d	920
	48.5 kg; n1	10118
	f: 55 kg; n1, 1 wk diseased	11785
	m: est 100 kg	2604
Dicerorhinus sumatrensis	23 kg; n1	4108
Diceros bicornis	est 22-34 kg	4521
	35 kg; r25-45	8798
	39.04 kg; n1 term emb	3830
	~40 kg	5762, 10158
Rhinoceros unicornis	33.75 kg; r33-34.5; n2	1820
	44.0 kg; r33.6-54.4; n2	155, 3656
	48.6 kg; n1, 2 d	7406
	56.8 kg; n1	6703
	m: 62.7 kg; r40-79; n9	153, 6181, 6185, 6194
	m: 63.5 kg; r59-68; n2	2674
	f: 68.6 kg; r55.7-81; n9	153, 6181, 6185, 6194
	70 kg	6190
	71.3 kg; r40-81; n20	6255
NEONATAL SIZE		
Dicerorhinus sumatrensis	914 mm; n1	4108
Rhinoceros unicornis	HB: 96.5-122 cm	6255
	99.0 cm; r96.5-101.6; n2	155
	105 cm; n1	6181

LITTER SIZE		
Ceratotherium simum	1 yg	4642a
	1 yg	5762, 8798, 10158
	1 yg	8514
	1 yg; n5 lt	5973
Dicerorhinus sumatrensis	1 yg	4108
Diceros bicomis	1	5762, 8798, 10158
	mode 1	2449
	1 yg	2599
	1 yg	9155
	1; n1 lt	5504
Rhinoceros unicomis	1 yg	1820
	1; n1 lt	6133
	1 yg; n1 lt	4197
	1 yg; n1 lt	6033
	1 ea; n24 lt	994
	1 yg ea; n27 lt	6194
	2 yg; with 1 f	3657
SEXUAL MATURITY		
Ceratotherium simum	f: 1st estrus: 3 yr	4642a
	f: 1st birth: 6.5-7 yr	8167
	f: 3 yr	4107
	m: 12 yr	8167
	f/m: 1st mate: ~5 yr	8973
	f/m: 4-5 yr	10158
	f/m: 7-10 yr	8798
Diceros bicomis	f: 1st conceive: 4.5 yr; n1	3829
	f: 1st birth: 4.75-5.25 yr	9592
	f: 1st birth: 6.42 yr	4831
	m: spermatogenesis: 7-8 yr	4831
	m: 1st mate: 5.5-6 yr	9592
	m: hold territory: 9 yr	4831
	m: 1st sire: 4.3 yr; n1	3829
	m: 6-7 yr	8798
	f/m: ~5-6 yr	5762
	f/m: 5-6 yr	5762
Rhinoceros unicomis	f: 1st estrus: 4 yr	6255
	f: 1st birth: 7.1 yr; r6-8	6251
	f: 4 yr	6190
	m: 7 yr	6190
CYCLE LENGTH		
Ceratotherium simum	estrus: 30 d	8167
Diceros bicomis	estrus: 6-7 d	4831
	18 d	4251
	26-30 d	2599
	28-30 d	11998
	30-35 d	3983
	35 d; sd7.7; r26-46; n10 cycles, 4 f	4831
Rhinoceros unicomis	estrus: 24 h	6181
	24 d; r21-33; n1 f	6255
	27-42 d	6251
	36-58 d	6181, 6185
	40-50 d; n2 yr, 1 f	10868
	40-50 d; n10 cycles; 1 78 d cycle, 1 126 d cycle	10867
	43 d; sd2; r41-45; n4 cycles, 2 f	5552
	45 d; r24-57; n3	1433
GESTATION		
Ceratotherium simum	>15 mo 3 wk	10118
	484 d; n1	9571
	~16 mo	5762, 10158
	16 mo	8167
	est 17 mo; ibi minus ppe	8973
	547 d; n1	8798
	18 mo	8514

	584 d; n1	11785
	~20 mo; n2	3062
Dicerorhinus sumatrensis	8 mo; questionable	4108
Diceros bicornis	431.67 d; r419-438; n3	4024
	15-17 mo; n5	4251
	458 d; n1	3983
	458.0 d; r454-463; n3	4521
	462 d; r446-478; n2	3828
	463.5 d; r462-465; n2	11998
	469 d	2599
	16-18 mo	5776
	530-550 d	2449
	540 d	9155
Rhinoceros unicornis	463 d; r462-464; n2	994
	464 d; n1	4197
	474 d; r471-477; n2	994
	478 d; r465-489; n17	994
	478.5 d; r462-489; n27	153, 6181, 6185, 6188, 6194
	479 d; r470-490; n3	994
	479 d; r462-491; n31	6255
	484 d; n1	901
	484 d; n1	7406
	486 d; n1	6033
	16 mo	1433
	488 d	10867, 10868
	18 mo	1820
	est 18.5-19 mo	155, 3656
LACTATION		
Ceratotherium simum	solid food: 1 wk	8798
	solid food: 2-3 wk	3062
	solid food: 2 mo	8167
	wean: ~1 yr	8167
Diceros bicornis	solid food: 9 d	8798
	solid food: 30 d	5504
	solid food: few wk	10158
	nurse: ~1 yr	10158
	nurse: 2 yr	9155
	wean: 2 yr	11697
	wean: >17.5 mo; n1f	3828
	wean: 1.5-2 yr	8798
Rhinoceros unicornis	solid food: 2-3 mo	6251
	solid food: 7 wk	6703
	wean: 12-18 mo	6251
	wean: 15 mo	6703
INTERLITTER INTERVAL		
Ceratotherium simum	21-22 mo; est from 6-7 mo ppe & gest	7204
	2-3 yr	8167
	3-5 yr	4246a
Diceros bicornis	2-4 yr	5762
	27.3 mo; r25-29; n3	3828
	2.5-3.25 yr	9592
	30.6 mo; r22-55; n13; 28 mo if drop >40 mo	4831
	~3 yr	10158
	44.5 mo; r20-89; n32; 32 mo if drop >40 mo	4831
Rhinoceros unicornis	2 yr	6188
	25.8 mo	9364
	median 34 mo; max 50; n50	6251

SEASONALITY

Ceratotherium simum	mate: Feb-May; Uganda	4642a
	mate: peaks Feb, Nov, few July-Sept; Zululand, S Africa	8168
	mate: peak Oct-Dec; Zululand, S Africa	8167
	birth: peak Mar-May; Zululand, S Africa	8167
	birth: peaks Mar, July; Zululand, S Africa	8168
	birth: yr rd; S Africa	10158
Diceros bicornis	estrus: Jan-Feb; Okawango, Angola	11697
	mate: peak est Mar-Apr, Sept-Nov; Kenya	5762
	mate: Aug-Dec; Kaokoveld, Namibia	5455
	mate: Nov-Dec; Okawango, Angola	11697
	mate: yr rd; Kenya	9155
	mate: yr rd; Tanzania	3827
	birth: Jan-May; Okawango, Angola	11697
	birth: peak Jan, June, low Mar-May; n Natal, S Africa	4831
	birth: Feb, May, July, Sept-Nov; n32; zoo	5818
	birth: peak June-July; Kenya	9592
	birth: Dec-Aug; Kaokoveld, Namibia	7204
	birth: yr rd; est peaks Jan, June; n Natal, S Africa	4831
	birth: yr rd; S Africa	10158
	birth: yr rd; Tanzania	5762
Rhinoceros unicornis	estrus: Aug; n1; zoo	1433
	mate: est Feb-Apr; India	3656
	birth: Oct; n1; zoo	10867
	birth: yr rd; Chitwan valley, Nepal	6251
	birth: yr rd; n36; zoo	6194